

- (b) $\vec{F}_{\text{perp}} = \vec{0}$
 (c) $W = -50$
- 49 (a) $\vec{F}_{\text{parallel}} = 3.846\vec{i} - 0.769\vec{j}$
 (b) $\vec{F}_{\text{perp}} = -3.846\vec{i} - 19.231\vec{j}$
 (c) $W = 20$
- 51 (a) $\vec{F}_{\text{parallel}} = \vec{0}$
 (b) $\vec{F}_{\text{perp}} = \vec{F}$
 (c) $W = 0$
- 53 No
- 55 (a) 2.24 m/sec
 (b) 1.89 m/sec
- 57 $10/\sqrt{30}$
- 59 Averages; weightings
 85.65%; class average
- 71 Can't take dot product of a scalar and a vector
- 73 Normal vector is $2\vec{i} + 3\vec{j} - \vec{k}$
- 75 $f(x, y) = (-1/3)x + (-2/3)y$
- 77 True
- 79 False
- 81 True
- 83 False
- 85 True

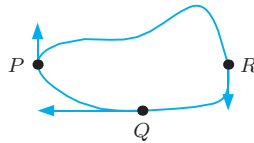
Section 13.4

- 1 $-\vec{i}$
- 3 $-\vec{i} + \vec{j} + \vec{k}$
- 5 $\vec{i} + 3\vec{j} + 7\vec{k}$
- 7 $7\vec{i} + \vec{j} + 4\vec{k}$
- 9 $-2\vec{k}$
- 11 $\vec{i} - \vec{j}$
- 13 $\vec{v} \times \vec{w} = -6\vec{i} + 7\vec{j} + 8\vec{k}$
 $\vec{w} \times \vec{v} = 6\vec{i} - 7\vec{j} - 8\vec{k}$
 $\vec{v} \times \vec{w} = -(\vec{w} \times \vec{v})$
- 15 $x - y - z = -3$
- 17 4
- 19 0
- 21 $4x + 26y + 14z = 0$
- 23 $3x - y - 2z = 0$
- 25 $4\vec{i} + 26\vec{j} + 14\vec{k}$
- 27 $4(x - 4) + 26(y - 5) + 14(z - 6) = 0$
- 29 (a) $\vec{u}$ and $-\vec{u}$ where
 $\vec{u} = \frac{12}{13}\vec{i} - \frac{4}{13}\vec{j} - \frac{3}{13}\vec{k}$
 (b) $\theta \approx 49.76^\circ$
 (c) $13/2$
 (d) $13/\sqrt{29}$
- 31 Parallel to the z -axis
- 33 $\sqrt{38}/3$
- 35 (a) Increases force
 (b) Ball moves down and to the left
- 37 (a) $\vec{P}_1\vec{P}_2 = \vec{P}_3\vec{P}_4 = 2\vec{i} + 4\vec{j} + 2\vec{k}$;
 $\vec{P}_1\vec{P}_3 = \vec{P}_2\vec{P}_4 = 3\vec{i}$
 (b) $\sqrt{180}$
- 51 $4\pi\vec{i}$
- 53 (a) $((u_2v_3 - u_3v_2)^2 + (u_3v_1 - u_1v_3)^2 + (u_1v_2 - u_2v_1)^2)^{1/2}$
 (b) $|u_1v_2 - u_2v_1|$
 (c) $m = (u_2v_3 - u_3v_2)/(u_2v_1 - u_1v_2)$,
 $n = (u_3v_1 - u_1v_3)/(u_2v_1 - u_1v_2)$
- 55 Parallel, not perpendicular
- 57 $\vec{v} = (8\vec{i} - 6\vec{j})/5$
- 59 False

- 61 True
- 63 True
- 65 True
- 67 False

Chapter 13 Review

- 1 Scalar; -1
- 3 -1
- 5 $\vec{a} = -2\vec{j}$, $\vec{b} = 3\vec{i}$, $\vec{c} = \vec{i} + \vec{j}$,
 $\vec{d} = 2\vec{j}$, $\vec{e} = \vec{i} - 2\vec{j}$, $\vec{f} = -3\vec{i} - \vec{j}$
- 7 $5\vec{i} + 30\vec{j}$
- 9 $3\sqrt{2}$
- 11 $3\vec{i} + 7\vec{j} - 4\vec{k}$
- 13 -3
- 15 $\vec{0}$
- 17 0
- 19 $\vec{0}$
- 21 $-5\vec{i} + 3\vec{j} + \vec{k}$ (multiples of)
- 23 (a) 4
 (b) $-4\vec{i} - 11\vec{j} - 17\vec{k}$
 (c) $3.64\vec{i} + 2.43\vec{j} - 2.43\vec{k}$
 (d) 79.0°
 (e) 0.784
 (f) $2\vec{i} - 2\vec{j} + \vec{k}$ (many answers possible)
 (g) $-4\vec{i} - 11\vec{j} - 17\vec{k}$
- 25 $\pm(-\vec{i} + \vec{j} - 2\vec{k})/\sqrt{6}$
- 27 $\vec{n} = 4\vec{i} + 6\vec{k}$
- 29 $-3\vec{i} + 4\vec{j}$
- 31 $\vec{u}$ and $\vec{w}$; $\vec{v}$ and $\vec{q}$.
- 33 $\vec{F}_{\text{parallel}} = \vec{F}$
 $\vec{F}_{\text{perp}} = \vec{0}$
 $W = -10$
- 35 $\vec{F}_{\text{parallel}} = -(6/5)\vec{i} + (8/5)\vec{j}$
 $\vec{F}_{\text{perp}} = (16/5)\vec{i} + (12/5)\vec{j}$
 $W = -10$
- 37 $\vec{F}_{\text{parallel}} = 2\vec{j}$
 $\vec{F}_{\text{perp}} = 5\vec{i}$
 $W = 6$
- 39 (a) True
 (b) False
 (c) False
 (d) True
 (e) True
 (f) False
- 41 548.6 km/hr
- 43

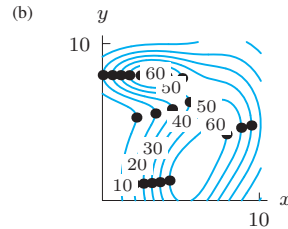
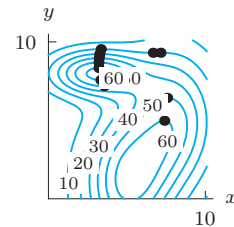


- 45 Parallel:
 $3\vec{i} + \sqrt{3}\vec{j}$ and $\sqrt{3}\vec{i} + \vec{j}$
 Perpendicular:
 $\sqrt{3}\vec{i} + \vec{j}$ and $\vec{i} - \sqrt{3}\vec{j}$
 $3\vec{i} + \sqrt{3}\vec{j}$ and $\vec{i} - \sqrt{3}\vec{j}$
- 47 2, 8
- 49 $2x - 3y + 7z = 19$
- 51 $\sqrt{6}/2$
- 53 (a) 1.5

- (b) $y = 1$
- 55 $9x - 16y + 12z = 5$
 0.23
- 57 $7.0710\vec{i} + 2.5882\vec{j} + 6.580\vec{k}$
- 59 228.43 newtons
 85.5° south of east
- 61 $|ywr - vzr + zus - xws + xvt - yut|$
- 63 (b) $(1, 1/\sqrt{3}, \sqrt{6}/6)$
 (c) 109.471°

Section 14.1

- 1 $f_x(3, 2) \approx -2/5$; $f_y(3, 2) \approx 3/5$
- 3 $-0.0493, -0.3660$
 $-0.0501, -0.3629$
- 5 $\partial P/\partial t$:
 dollars/month
 Rate of change in payments with time
 negative
 $\partial P/\partial r$:
 dollars/percentage point
 Rate of change in payments with interest rate
 positive
- 7 (a) Payment \$376.59/mo at 1% for 24 mos
 (b) 4.7¢ extra/mo for \$1 increase
 (c) Approx \$44.83 increase for 1% interest increase
- 9 (a) Negative
 (b) Positive
- 11 $f_x > 0$, $f_y < 0$
- 13 $f_x < 0$, $f_y > 0$
- 15 $f_T(5, 20) \approx 1.2^\circ\text{F}/^\circ\text{F}$
- 17 Positive, Negative, 10, 2, -4
- 19 $\partial Q/\partial b < 0$
 $\partial Q/\partial c > 0$
- 21 -1.5 and -1.22
- 23 (a)

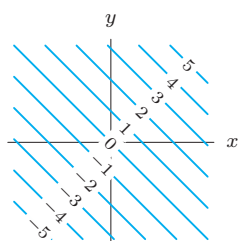


- 25 (a) Negative
 (b) Positive
- 27 (a) 2.5, 0.02
 (b) 3.33, 0.02
 (c) 3.33, 0.02
- 29 (i)(c); (ii)(a)
- 31 -2.5

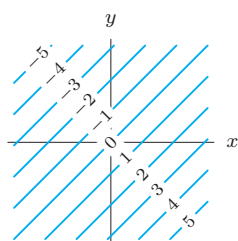
33

T ($^{\circ}\text{C}$)	w (gm/m^3)		
	0.1	0.2	0.3
	10	1300	900
	20	800	800
	30	800	700

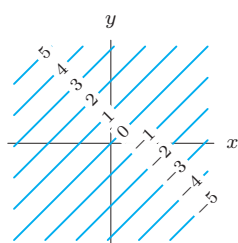
35 (a)



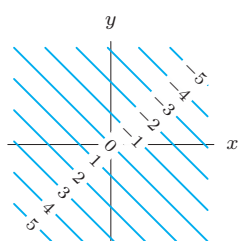
(b)



(c)



(d)



37 $f_y = 0$

39 $f(x, y) = 4x - y$

41 False

43 False

45 True

47 False

49 False

Section 14.2

1 (a) 7.01

(b) 7

3 $f_x(1, 2) = 15$,

$f_y(1, 2) = -5$

5 $\partial z / \partial x = \frac{14x+7}{(x^2+x-y)^{-6}}$

$\partial z / \partial y = -7(x^2 + x - y)^6$

7 $f_x = 0.6/x$

$f_y = 0.4/y$

9 $2xy + 10x^4y$

11 $-2\pi r/T^2$

13 $e^{\sqrt{xy}}(1 + \sqrt{xy}/2)$

15 g

17 $(a + b)/2$

19 $2B/u_0$

21 $2mv/r$

23 Gm_1/r^2

25 $-e^{-x^2/a^2}(a^2 - 2x^2)/a^4$

27 $v_0 + at$

29 $-\pi r^{3/2}/(M\sqrt{GM})$

31 $(15x^2y - 3y^2)\cos(5x^3y - 3xy^2)$

33 $\partial F / \partial L = \frac{3}{2}\sqrt{K/L}$

35 $\epsilon_0 E$

37 $c\gamma a_1 b_1 K^{b_1-1}(a_1 K^{b_1} + a_2 L^{b_2})^{\gamma-1}$

39 13.6

41 (a) 3.3, 2.5

(b) 4.1, 2.1

(c) 4, 2

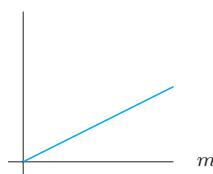
43 (a) $a < 3$

(b) Up

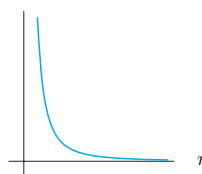
45 (a) $\partial g / \partial m = G/r^2$

$\partial g / \partial r = -2Gm/r^3$

(b) g



g



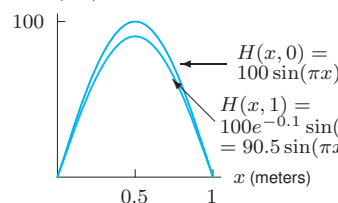
r

47 (a) $c^2((1 - v^2/c^2)^{-1/2} - 1)$; positive

(b) $mv(1 - v^2/c^2)^{-3/2}$; positive

49 (a)

temp ($^{\circ}\text{C}$)



(b) $254.2e^{-0.1t}$ $^{\circ}\text{C/m}$

(c) $-254.2e^{-0.1t}$ $^{\circ}\text{C/m}$

(d) $-10e^{-0.1t} \sin(\pi x)$ $^{\circ}\text{C/sec}$

51 $f(x, y) = x^4y^2 - 3xy^4 + C$

53 Derivative is limit, not a difference quotient
Counterexample: $f(x, y) = x^2 + y^2$

55 $f(x, y) = xy, g(x, y) = xy + y^2$

57 True

59 True

61 False

63 False

65 (d)

Section 14.3

1 $z = ex$

3 $z = 6y - 9$

5 $z = -4 + 2x + 4y$

7 $z = -36x - 24y + 148$

9 $df = y \cos(xy) dx + x \cos(xy) dy$

11 $dz = -e^{-x} \cos(y) dx - e^{-x} \sin(y) dy$

13 $dg = 4 dx$

15 $dP \approx 2.395 dK + 0.008 dL$

17 (a) Dollars/Square foot

(b) Larger plots at same distance \$3/ft² more

(c) Dollars/Foot

(d) Farther from beach but same area \$2/ft less

(e) 998 ft²

19 (b) $f(x, y) \approx$

$0.3345 - 0.33(x - 1) - 0.15(y - 2)$

(c) $f(x, y) \approx 0.3345 -$

$0.3345(x - 1) - 0.1531(y - 2)$

21 376

23 $df = \frac{1}{3}dx + 2dy$

$f(1.04, 1.98) \approx 2.973$

25 136.09 $^{\circ}\text{C}$

27 $P(r, L) \approx$

$80 + 2.5(r - 8) + 0.02(L - 4000)$

$P(r, L) \approx$

$120 + 3.33(r - 8) + 0.02(L - 6000)$

$P(r, L) \approx$

$160 + 3.33(r - 13) + 0.02(L - 7000)$

31 (a) $nRT/(V - nb) - n^2a/V^2$

(b) $\Delta P \approx \frac{nR}{(V_0 - nb)} \Delta T + \frac{(2n^2a/V_0^3 - nRT_0/((V_0 - nb)^2)) \Delta V}{}$

33 (a) $d\rho = -\beta\rho dT$

(b) 0.00015, $\beta \approx 0.0005$

35 $-43200\Delta t$

Slow if $\Delta t > 0$; fast if $\Delta t < 0$

37 $z = f(3, 4) + f_x(3, 4)(x - 3) + f_y(3, 4)(y - 4)$

39 Equation not linear

41 sphere of radius 3 centered at the origin

43 True

45 False

47 True

49 False

Section 14.4

1 $(\frac{15}{2}x^4)\vec{i} - (\frac{24}{7}y^5)\vec{j}$

3 $2m\vec{i} + 2n\vec{j}$

5 $\left(\frac{5\alpha}{\sqrt{5\alpha^2 + \beta}}\right)\vec{i} + \left(\frac{1}{2\sqrt{5\alpha^2 + \beta}}\right)\vec{j}$

7 $\nabla z = e^y\vec{i} + e^y(1 + x + y)\vec{j}$